Module: Machine Learning for Engineers

ASSESSMENT: ML-Based Visual Quality Inspection System

1. Objective

Design, train, and deploy a machine-learning model that classifies objects as PASS or FAIL based on camera images, as part of an automated quality inspection system.

This task focuses on ML for engineering decision-making.

2. System description

The inspection system operates as follows:

- Object is placed in a fixed inspection area
- Camera captures an image
- ML model classifies object quality
- System outputs a decision: PASS or FAIL

Choose one from the following examples of acceptable quality checks. Ground truth must be clearly defined and verifiable.

- Defective vs non-defective object
- Correct vs incorrect orientation
- Correct vs incorrect size or shape

Note: Model complexity is not rewarded over correctness and clarity.

3. Deliverables

Demonstration

- Working inspection pipeline
- Live or recorded test on new samples

Short report (2 pages)

Must include:

- Problem definition & ground truth
- Model design & training including pseudo-code description of the training loop
- Quantitative results
- Failure cases & limitations

Code: Labcode link

4. Rubric (100%)

Criterion	Weight	Notes
Demonstration	50%	Performance evaluation. Split into 5 tests and 10% for each correct output.
Report	40%	10% for each of the components (mentioned in Deliverables.
Code	10%	Data, code, and ReadMe. Commits over time.